

Human-Action Driven UI Architecture Proposal

Product Direction: Human-Action Organization | Date: September 2, 2026

1. Executive Product Decision: Human-Action Centric UI

Decision: The 'Search-First' / Command-Palette model has been rejected. Instead, PrepBench will be structured strictly around **Clear Human Actions** (verbs describing what a user wants to do: *Practice, Learn, Review, Configure*).

Why Human Actions Win:

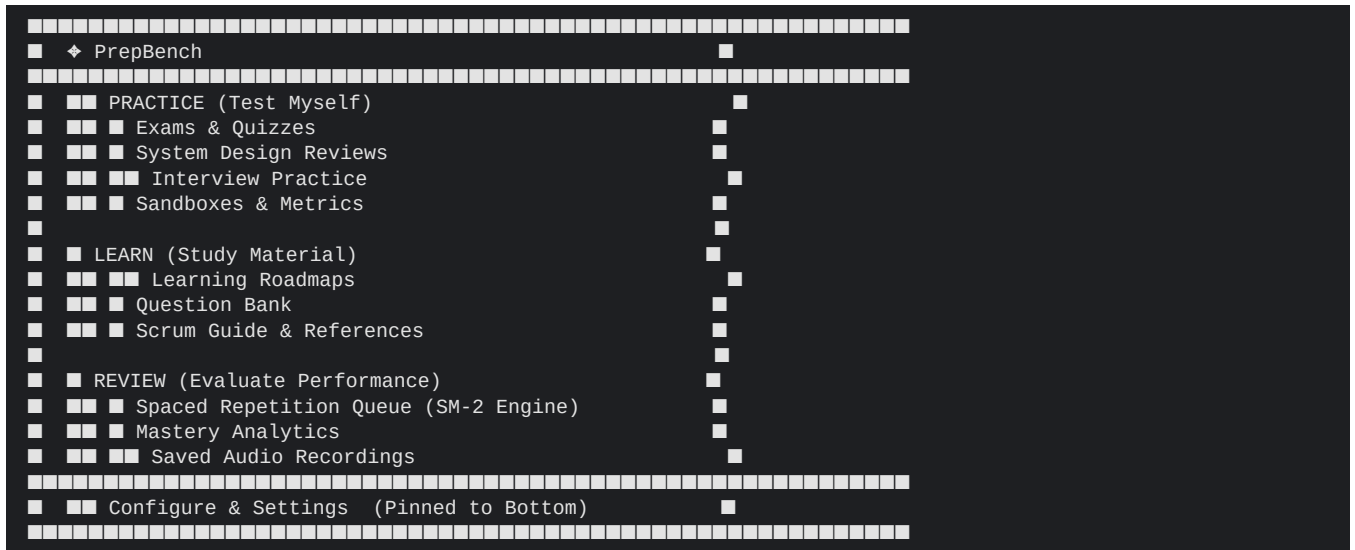
- 1. **Zero Cognitive Friction:** Users do not need to memorize search terms or suffer from empty text box paralysis.
- 2. **Predictable Navigation:** Every feature has an intuitive home based on human intent.
- 3. **Unlimited Subject Scaling:** Subjects (Scrum, Databricks, System Design, AI) operate as clean options inside each action hub, preventing sidebar menu bloat forever.

2. The 4 Human Action Navigation Hubs

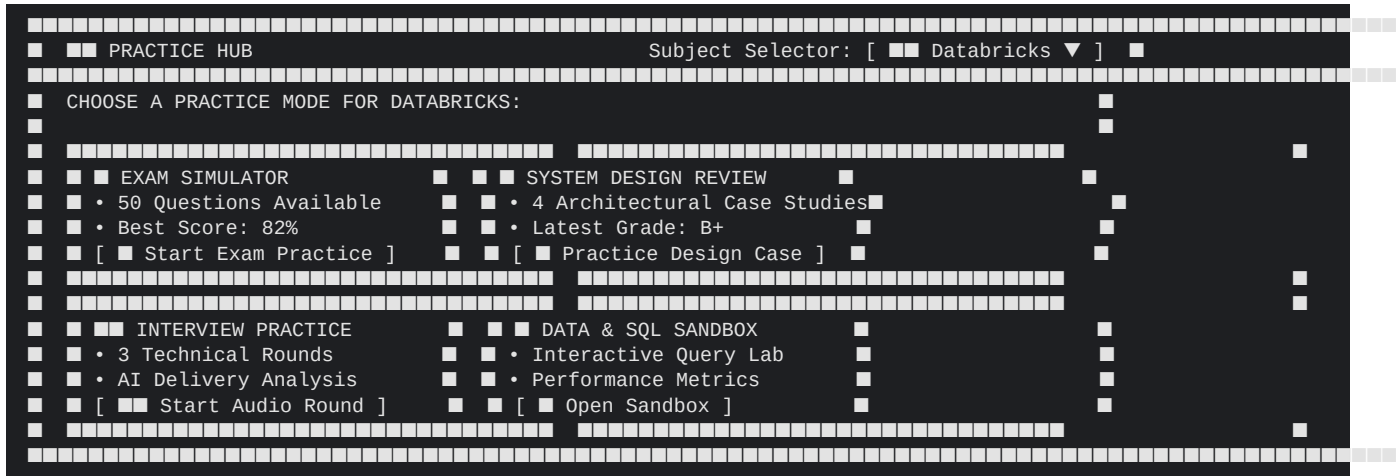
Human Action Hub	User Intent (What the user wants to do)	Included Features & Modules
■■ 1. PRACTICE	'I want to test myself & do active exercises.'	• Multiple-Choice Exams • System Design Reviews • Interview Audio Practice • Sandboxes & Interactive Labs
■ 2. LEARN	'I want to study material & acquire knowledge.'	• Sequential Learning Roadmaps • Question & Flashcard Bank • Scrum Guide & Concept Reference Docs
■ 3. REVIEW	'I want to evaluate outputs & review memory.'	• Spaced Repetition Queue (SM-2 Engine) • Unified Mastery Analytics • Saved Interview Audio Recordings
■■ 4. CONFIGURE	'I want to manage settings & platform options.'	• App Preferences & Themes • AI Key Management (Gemini) • Database Reset & Backup

3. Human-Action Navigation Wireframe

A. Clean Sidebar Structure (Grouped by Human Actions)



B. Practice Hub Dashboard Surface



4. Surfacing the Backend SM-2 Spaced Repetition Engine

The backend already contains `sm2_service.py` and the SpacedRepetition database model. Under the Human-Action architecture, this engine gets a dedicated home under **REVIEW** → **Spaced Repetition Queue**.

How Spaced Repetition Works for the User:

- 1. As the user completes practice exams and quizzes, `sm2_service.py` tracks question difficulty and calculate next review intervals.
- 2. Under **REVIEW** → **Spaced Repetition Queue**, the user sees: *'14 Questions Due for Memory Review Today'*.
- 3. One click launches a 5-minute review session targeting only items scheduled by the memory decay algorithm.